

Python for Beginners

A simple guide to learning Python step by step.

1. What is programming?

Programming means giving instructions to a computer. A program is a set of steps the computer follows in order. Example: `print("Hello, world!")`

2. Variables

Variables store information. You can think of them as labeled boxes that hold data. Example: `name = "Ada"`

3. Input and output

Use `print()` to show output and `input()` to ask the user for information.

4. Numbers and math

Python can work with numbers using `+`, `-`, `*`, and `/`. Example: `subtotal = price * quantity`

5. Decisions with if

The `if` statement lets your program make decisions. It can choose one path or another.

6. Repeating with loops

Loops let you repeat code. Example: `for i in range(3): print("Hello")`

7. Functions

Functions let you organize code into reusable blocks. They help keep programs clear and simple.

8. Understanding your receipt generator

Your receipt program uses input, variables, loops, and math to calculate totals and tax.

9. Small practice exercise

Try this example: `name = input("What is your name? ") age = int(input("How old are you? "))`

10. Suggested learning order

Start with variables, input and output, numbers, if statements, loops, and functions.

Final advice

The best way to learn Python is to write small programs, test them often, and fix errors one at a time.